

BioSpecInfo

Software Bill of Materials (SBOM)

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Application version	bsi-v197
Commit	8b0f183a8818921b1d69285378a786e6e108df60
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Machine-readable format	<code>../evidence/sbom.cdx.json</code> — CycloneDX 1.5

The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) _{RDKit / Greg Landrum}	<code>RDKit_minimal.js</code>	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) _{RDKit / Greg Landrum}	<code>RDKit_minimal.wasm</code>	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js _{mrdoob e collaboratori}	<code>three.min.js</code>	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass _{mrdoob e collaboratori}	<code>three_bloom.js</code>	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader _{mrdoob e collaboratori}	<code>gltf_loader.js</code>	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js _{University of Pittsburgh / David Koes}	<code>3Dmol-min.js</code>	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).

Component	Version/File	Licence	Size	Role
SmilesDrawer _{Daniel Probst}	smiles-drawer.min.js	MIT	246 kB	2D structure drawing from SMILES.
PDF.js (build legacy) _{Mozilla Foundation}	vendor/pdfjs/pdf.min.mjs	Apache-2.0	511 kB	Opening PDF documents in the spectrum reader: every page drawn and its text layer. It is the legacy build, because the default one uses very recent language features and breaks on browsers that do not have them yet.
PDF.js — worker _{Mozilla Foundation}	vendor/pdfjs/pdf.worker.min.mjs	Apache-2.0	1290 kB	The PDF decoding work, off the main thread: a complex page does not freeze the interface.
sql.js (JS glue) _{sql.js / SQLite}	lib/sql-wasm.js	MIT	45 kB	In-browser SQLite database for the data exercises.
SQLite (WebAssembly) _{SQLite Consortium}	lib/sql-wasm.wasm	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) _{CERN Open Data Portal}	lib/dimuon.js	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.

Digests

File	SHA-256
RDKit_minimal.js	58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9
RDKit_minimal.wasm	e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8
three.min.js	9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2
three_bloom.js	42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090
gltf_loader.js	5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6
3Dmol-min.js	06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff
smiles-drawer.min.js	955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2
vendor/pdfjs/pdf.min.mjs	bccc24ea711db8e44503629519904a5292d73b9daaa214bbe7cdcc282b0f4259
vendor/pdfjs/pdf.worker.min.mjs	145d2dd3ab0c86151011dba95acfa2d5336e2accd59388ea43dbee0efddeaec6
lib/sql-wasm.js	77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f
lib/sql-wasm.wasm	438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed
lib/dimuon.js	61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d

2. Code written for this project

File	Size	Role
<code>index.html</code>	4692 kB	Main application: interface, teaching sections, chemical data.
<code>bsi-ai-hub.js</code>	422 kB	«Spectra» AI agent: agentic loop, tools, provider management.
<code>bsi-spettri.js</code>	35 kB	IR/NMR spectral prediction engine over the molecular graph.
<code>bsi-cheminfo.js</code>	125 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
<code>bsi-prettsch.js</code>	37 kB	The NMR estimation tables transcribed in full from Prettsch-Bühlmann-Badertscher, 4th ed.: ^{13}C and ^1H benzenes, ethylenes, alkanes, alkynes, ^{13}C aliphatics, steric corrections, $^1\text{J}(\text{C,H})$. Numbers only, no code.
<code>bsi-nmr.js</code>	57 kB	Per-atom ^1H and ^{13}C NMR prediction: applies the bsi-prettsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.
<code>bsi-geom3d.js</code>	46 kB	3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link.
<code>bsi-nmr2d.js</code>	21 kB	Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions.
<code>bsi-moldraw.js</code>	44 kB	Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports.
<code>bsi-documento.js</code>	23 kB	Opening of any document — PDF, images, text, Word, OpenDocument, PowerPoint, Excel — with an explicit list of what could NOT be read.
<code>bsi-quesito.js</code>	19 kB	Recognition of spectroscopic data in a problem's text, with each datum's provenance, and step-by-step working through bsi-elucida.js.
<code>bsi-spettrolettore.js</code>	79 kB	Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses.
<code>bsi-mol3d.js</code>	42 kB	3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.
<code>bsi-lingue.js</code>	159 kB	Internationalisation engine: fourteen languages, writing direction included.
<code>bsi-molingue.js</code>	54 kB	Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.
<code>bsi-pannelli-lingua.js</code>	66 kB	Panels for languages and chemical languages.
<code>sw.js</code>	5 kB	Service Worker: offline operation and network strategy.
<code>rdkit_lab.html</code>	260 kB	Cheminformatics laboratory.
<code>astro.html</code>	2660 kB	Astrochemistry module.

File	Size	Role
chimorga.html	194 kB	Organic chemistry module.
proxy/spectra-proxy.js	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
index.html	c4b24c7a5c7df24f4e85e341d3ff5fa0b45aeb92520c45525c3565a83724670c
bsi-ai-hub.js	985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476
bsi-spettri.js	be757f15a70f884e3374d401596e2dfd18bcc852cafb4584920e2b2d24eb3c9
bsi-cheminfo.js	59a3111b3b9436add5376fde1be6f90ded3dc08c095cebd04da50c7f8268bbc
bsi-pretzsch.js	27f727240363579fba3dba50ab3b567fbcdb26a8a9a06c0e43968b195c479db3
bsi-nmr.js	881917dbb141dfcbab9a25b9c518eaa4a8c1c52dc1020cbac90985a1cf859b72
bsi-geom3d.js	a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054
bsi-nmr2d.js	a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78
bsi-moldraw.js	45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e
bsi-documento.js	0e9ad6043a396652549b30879764cc401f8b7bd25689b5bb5f37cf26f7d7d8d9
bsi-quesito.js	6f686abcfdc3959d8c865a3e6ee7129224de52992ab1ba3da07372a986769cdf
bsi-spettrolettore.js	cd19768291586d099acf0fefe06d34417644a83b1b952d0943ae3a16c491f4
bsi-mol3d.js	93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893
bsi-lingue.js	588beac68e5ea9fcfd430a3b56290eb61df509a41c507aa24ba8c146a809431
bsi-molingue.js	e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb
bsi-pannelli-lingua.js	4af94d70ef104cf21be18ff7fbc7a2248c56e376e2ae48e8cac83878f4554f93
sw.js	862b7e801bc095785b851687f3ccb52201736d8476faef71b268259489e8c5d2
rdkit_lab.html	066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad
astro.html	3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d
chimorga.html	047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d
proxy/spectra-proxy.js	808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
playwright-core	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage` , or on the user's own proxy (see `proxy/`).

5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md` .

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